Centered Pell norms and shared positivity

A degree-fourteen consistency equation in twenty-two integers

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Abstract

We construct an explicit sum of ten squares in twenty-two unrestricted integer unknowns, of exact total degree fourteen. Its integer solvability is equivalent to finite ZFC refutability through the repository's specified native/compiler calibration. Centering an auxiliary Pell root and applying an elementary row operation permit the last coordinate elimination. A second norm transformation, justified by a scale that never vanishes, gives a degree-ten equation in twenty-four integers. The intermediate degree-twelve equation in twenty-three integers uses only 122 binary operations. These constructions combine the signed-power bootstrap, a calibrated determinant, and a shared strict-positivity certificate. Complete leading forms and exact polynomial identity checks certify the new endpoints. Companions have thirty-one integer unknowns at degree six and twenty-six at degree eight. Separately, adjacent-coefficient Pell descent removes an unnecessary divisibility factor from the eleven-positive-variable form, lowering its degree from 325 to 277 and its circuit from 113 to 111 binary operations. A twelve-positive-variable companion has degree 78 in 103 operations. We also prove an index-elimination lemma for a different CRT proof-certificate architecture; that lemma is not yet compiled into another consistency polynomial. All fixed coefficient integers, polynomial circuits, proof dependencies, checks, and source pins accompany this article.

Contents

1 Outcome, comparison, and the exact claim **2**

2 The main equation: twenty-three unrestricted integers **3**

3 Centered norms: twenty-four and twenty-two integers **7**

4 Adjacent Pell descent and the eleven-positive-variable equation **10**

5 A twelve-positive-variable companion of degree 78 **14**

6 A separate compiler improvement: the modulus supplies the index **15**

7 Reproducibility and interpretation of the checks **17**

8 Further research questions **18**

1 Outcome, comparison, and the exact claim

The smallest new low-degree construction is P_{22} in Section 3: twenty-two unrestricted integer unknowns, ten squared residuals, exact degree fourteen, and 123 binary arithmetic operations. It is obtained by explicit transformations of the completely displayed degree-twelve polynomial P_{23} in Section 2. A degree-ten companion P_{24} retains twenty-four unknowns. All coefficient integers are materialized in the archive. Their domains are respectively $\mathbb{Z}^{22}$, $\mathbb{Z}^{23}$ and $\mathbb{Z}^{24}$, with no sign conditions left outside the equations.

The repository was inspected at commit `2605669a532342f153b13eb6b4a0af4b54f5fa37`. Its catalogue includes sources 31–75 and the reconciled wave-10 frontier. Comparisons in this article refer to that pin, not to the older baseline used by each incoming package. The source numbers below refer to directories in `Computability/DiophantineConsistency/con-zfc-diophantine-forms`.

Domain and objective	Prior inspected construction	This contribution
Low-degree integer tradeoffs	26 integers, degree 16 (68); also 24 integers, degree 1936 in the catalogue	22 integers, degree 14; 23 integers, degree 12; 24 integers, degree 10
Integer sextic	32 integers (69, 71)	31 integers, eighteen residuals
Integer octic	27 integers (69, 71)	26 integers, fourteen residuals
Eleven positive integers	Degree 325, 113 operations (70)	Degree 277, 111 operations
Twelve positive integers	Degree 82, 102 operations (69)	Degree 78, 103 operations

Each new first-row construction improves both coordinates of the prior 26-integer/degree-16 result. The 22- and 23-integer equations improve both coordinates of the older 24-integer/degree-1936 result as well; P_{24} keeps its variable count and lowers its degree. The last row trades one extra arithmetic operation for four degrees. The nine-positive-unknown and ten-integer-variable high-degree records are unchanged. Thus there is no assertion of an absolute smallest equation, or of a uniform improvement in every complexity measure. In particular, operation counts do not measure bit complexity: some coefficient integers have more than a million bits.

1.1 The consistency statement and its proof boundary

Let $\mathrm{Ref}_{\mathrm{ZFC}}$ mean existence of a finite refutation in the recursively specified calculus used by the pinned K215 calibration. The target is the internal mathematical assertion

$$\mathrm{ZFC} \vdash (\neg\exists \mathbf{x} \in \mathcal{D}\ P(\mathbf{x}) = 0) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{1}$$

Here $(P, \mathcal{D})$ can be any of the seven delivered endpoint/domain pairs, in particular $(P_{22}, \mathbb{Z}^{22})$. Consistency is syntactic: Con(ZFC) abbreviates absence of a finite refutation. No assumption of consistency or arithmetic soundness is used in the reductions.

The new coordinate transformations, positivity lemmas and Pell descent are proved below in ordinary mathematics formalizable in ZFC. The final interpretation uses the repository's stated native/compiler theorem identifying its particular fixed-coefficient source with $\mathrm{Ref}_{\mathrm{ZFC}}$, and its canonical-witness theorem. Those dependencies are included and pinned, rather than replaced by finite tests. In particular, the integer construction depends on sources 64, 71 and 75, and the positive constructions on sources 44, 61, 69 and 70.

This is an ordinary mathematical contribution with executable algebra audits. It is not a newly compiled Lean/Rocq certificate of (1). The repository's `OBJECT_ZFC.md` distinguishes external Lean equivalence from an actual derivation in its encoded ZFC calculus; that formalization obligation is not silently declared closed here. Likewise, `NATIVE_CALIBRATION52.md` distinguishes the native arithmetic from its complete formal machine/tableau bridge. The archive retains both status documents. If one requires a new end-to-end formal derivation file rather than the inherited ordinary calibration theorem, that stronger deliverable remains open.

1.2 What is new, and what is inherited

The cone, the nearest-square positivity lemma, and the adjacent-coefficient Pell descent already occur separately in the latest incoming packages. We do not claim those ingredients as new. The contributions here are their specific compositions, the additional eliminations those compositions permit, the resulting literal smaller equations, and the proofs and exact audits of the new endpoints. In particular, the shared guard makes elimination of both D and V affordable at degree twelve. The centered auxiliary norm, the nonzero-scale replacement of the main norm, and the strong-norm row operation then yield the 24-integer/degree-10 and 22-integer/degree-14 endpoints. These last steps go beyond simply substituting monically defined coordinates. The CRT index substitution in Section 6 is a separate newly developed compiler-stage improvement. No global historical priority claim is made for any of these mechanisms.

2 The main equation: twenty-three unrestricted integers

2.1 Fixed numerals and the complete polynomial

Use the coherent source-34 tuple, named `mixed-parity-unrounded-1064921`, together with source 71's two determinant scales:

$$\beta = 2^{1064921} - 1, \qquad A_0 = 2^{1113}, \qquad B_0 = 2^{666104}.$$

The remaining three native numerals are K_0, M_C, M_F. Their complete integer bytes, their hashes and their native binding are in `integer/coefficients/BINDING.json` and its six payloads. M_F is the already shifted source mask. These symbols are fixed coefficients, not unknowns. They are the exact integers of the pinned source, not arbitrary integers satisfying some inequalities.

The twenty-three unknowns, all unrestricted integers, are

$$J, F, \alpha, f, h, r, v, \tau, \eta, y, \gamma, o, j, E, c, a, I, u, Z, t, p_1, p_2, p_3. \tag{2}$$

In the data files, v, t, p_1, p_2, p_3 are named `v_scale`, `pair_root`, and `pair_pos1`, `pair_pos2`, `pair_pos3`. Define only polynomial abbreviations:

$$\begin{aligned}
q &= \beta J + 1, & C &= q - F - \alpha, & Y &= a - E,\\
H &= 4a + 3, & \Delta &= a^2 + H, & X &= 4qu^2,\\
D &= X + ac + \gamma H, & V &= fo - c, & k &= 1 - V + jc + hE,\\
\zeta &= k - \eta, & \Pi &= 1 + p_1^2 + p_2^2 + p_3(p_3 + 1).
\end{aligned}$$

Also put

$$\begin{aligned}
R &= \big((q-1)(q-F) + \alpha\big)(q^2 - 1) + (M_C + qM_F)J,\\
G &= (A_0F)(B_0C) - (q - A_0F - B_0C)^2.
\end{aligned}$$

The eleven residuals are

$$\begin{aligned}
r_1 &= Y - (2v+1)q^3,\\
r_2 &= E - XY,\\
r_3 &= c - \eta - kY,\\
r_4 &= \tau^2 + (2\tau - k)E(c-\eta) - 1,\\
r_5 &= D^2 - \Delta c^2 - 1,\\
r_6 &= jc - V - R,\\
r_7 &= (K_0 q + X)C - qF - Z(q-1),\\
r_8 &= I - (2r+1)c^2,\\
r_9 &= I^2 - \Delta(f^2-1),\\
r_{10} &= y^2 + I^2(V^2 - y^2) - 1,\\
r_{11} &= G\eta\zeta - (t^2 - G - \eta\zeta + 1)^2 - \Pi.
\end{aligned} \tag{3}$$

The proposed single Diophantine equation is

$$\boxed{P_{23} = \sum_{\nu=1}^{11} r_\nu^2 = 0, \qquad \mathbf{x} \in \mathbb{Z}^{23}.} \tag{4}$$

All abbreviations are to be substituted; none introduces another existential coordinate. Every exponent involving an unknown is fixed. The companion JSON circuit evaluates precisely this polynomial in 122 live binary arithmetic operations.

Theorem 2.1 (The 23-integer reduction)**.** *The polynomial* (4) *has exact total degree twelve. Its integer solvability is equivalent to the existence of a finite ZFC refutation through the pinned native/compiler calibration of sources 64, 71 and 75. Consequently its unsolvability has the consistency interpretation* (1)*, with the proof boundary stated in Section 1.*

We prove the new parts in detail, and specify the inherited interface at the point where it is used.

2.2 One certificate for two positive quantities

For integers b_1, b_2, b_3 put

$$\mathrm{Pos}(b_1, b_2, b_3) = 1 + b_1^2 + b_2^2 + b_3(b_3+1).$$

This is always strictly positive. Conversely every positive integer n has this form: $4n-3$ is positive and congruent to 1 or 5 modulo 8, so Legendre's three-square theorem applies. Modulo 4, exactly one of the three squares is odd. Writing its root as $2b_3+1$ and the other two as $2b_1, 2b_2$ proves the assertion.

Lemma 2.2 (Shared positivity, source 68)**.** *For integers x, y, one has $x > 0$ and $y > 0$ if and only if there are integers t, b_1, b_2, b_3 such that*

$$xy - (t^2 - x - y + 1)^2 = \mathrm{Pos}(b_1, b_2, b_3). \tag{5}$$

Proof. If the left side is positive, then $xy > 0$. Both numbers therefore have the same nonzero sign. They cannot both be negative: writing $x = -a, y = -b$, with $a, b \geq 1$, makes the subtracted square at least $(a + b + 1)^2 > ab$.

Conversely suppose $x, y \geq 1$ and set $N = x + y - 1$. Then $xy - N = (x - 1)(y - 1) \geq 0$. Let $m = \lfloor \sqrt{N} \rfloor$. Choose $t = m$ if $N - m^2 \leq m$, and $t = m + 1$ otherwise. If N is a square the error is zero. Otherwise $|t^2 - N| \leq m$ and $m^2 < N$, so $(t^2 - N)^2 < N \leq xy$. The resulting positive difference has a Pos representation. The square root and floor select witnesses in the proof; they are not operations in the polynomial. □

Apply this to $x = G$, $y = \eta\zeta$. The final row in (3) therefore enforces exactly

$$G > 0, \qquad \eta\zeta > 0.$$

It uses four private coordinates instead of the six needed by two independent three-square representations. The equivalence holds for every integer assignment, including assignments with a negative radix.

2.3 The determinant and its canonical calibration

Set $U = A_0F$ and $W = B_0C$. If $G = UW - (q - U - W)^2 > 0$, then U, W have one common nonzero sign and

$$|q - U - W| < \sqrt{UW} \leq \frac{|U| + |W|}{2}.$$

It follows that q has that common sign and

$$|q| > \frac{A_0|F| + B_0|C|}{2} \geq |F| + |C|.$$

Thus F, C and $\alpha = q - F - C$ all have the same strict sign as q. In particular $F\alpha > 0$ and $FC > 0$. A new zero can therefore restore the two older native sign guards by independent Pos representations. This implication is elementary and uses no native compiler theorem.

The converse uses canonical witnesses; the determinant is not intended to contain every positive source assignment. Source 71 proves that the canonical native accepting words for this exact tuple satisfy

$$\frac{q}{2} \leq A_0F < \frac{4q}{5}, \qquad \frac{q}{2} \leq B_0C < \frac{3q}{5}, \qquad q > 0. \tag{6}$$

Here is the numerical content of that calibration, so its dependence is auditable. The native radix is $\rho = 2^{121}$ and the cell length is 8801. Every occupied canonical cell has top C digit 1 at position 3296. Its top F digit at position 8791 is

$$c_0 = 105534389824824194647382298625, \qquad 2^{96} \leq c_0 < c_0 + 1 \leq 3 \cdot 2^{95}.$$

The carry-free block bounds are

$$\rho^{3296} \leq C_j < (1 + 1/\rho)\rho^{3296},$$
$$c_0\rho^{8791} \leq F_j < (c_0 + 1)\rho^{8791}.$$

The exponent identities

$$121 \cdot 8791 + 1113 = 1064921 - 97, \qquad 121 \cdot 3296 + 666104 = 1064921 - 1$$

give (6) after summing the canonical cells. The full digit-support and carry proof is retained as `integer/upstream/DETERMINANT_GUARD.md`.

To see that these bounds imply $G > 0$, write $s = q/2$, $A_0F = s + x$, $B_0C = s + y$. Then $0 \le x < 3s/5$ and $0 \le y < s/5$, whence

$$G = s^2 + s(x + y) - (x^2 + y^2 + xy) > \frac{12}{25}s^2 > 0.$$

Source 71's bounds explicitly allow the lower dummy digits used to tune the rank. Source 75's signed-bootstrap change modifies only its square witness and cleared transport quotient, leaving J, F, C, α unchanged. Thus the canonical determinant and the signed bootstrap apply to the same family of witnesses. This compatibility is needed for completeness and was checked separately.

2.4 Restoring the signed source

Source 75 supplies a forty-integer-variable source with the arithmetic constraints $X = 4qu^2$ and

$$(K_0q + X)C - qF - Z(q - 1) = 0.$$

Its signed Pell argument recovers $X = 2^R > 0$, $q > 0$ and the odd rank R, without assuming these signs at the start. The transport equation then gives $q \mid Z$, restoring the older quotient. Canonical completeness has $q = 2^s$, with s and R odd and $R - s - 2 \ge 0$, so $u = 2^{(R-s-2)/2}$ exists. This inherited existential source reduction is retained in `integer/upstream/SIGNED_BOOTSTRAP.md`.

To restore its literal arithmetic from a zero of P_{23}, use

$$\begin{aligned} L &= q^2 - 1, & N &= q^3, & P &= k + E(c - \eta), \\ C_2 &= c^2, & F_2 &= f^2, & g &= y^2 - V^2, & B_{\text{aux}} &= I^2, \end{aligned}$$

together with the abbreviations already displayed and the root shear

$$\tau_{\text{old}} = \tau + E(c - \eta).$$

All these restored coordinates are integer polynomials. The shear is exact because, writing $P - k = E(c - \eta)$,

$$(\tau + P - k)^2 - P^2 + Pk - 1 = \tau^2 + (2\tau - k)E(c - \eta) - 1.$$

Twelve of the signed source's twenty-two arithmetic rows become zero polynomial identities; the other ten become $r_1, \dots, r_{10}$ up to sign. The independent audit verifies this entire identity list.

The last row restores $G > 0$ and $\eta\zeta > 0$ by Lemma 2.2. The determinant then restores the old $F\alpha$ and FC sign guards. This gives a signed-source zero and therefore a finite refutation by its inherited soundness theorem.

Conversely, a finite refutation supplies canonical signed-source witnesses. The determinant is positive on them by (6), and $\eta\zeta > 0$. Select the four shared positivity witnesses, apply the inverse root shear, and project away the monically defined coordinates. Every residual in (3) vanishes. This proves the two existence implications. The determinant replacement is an existence equivalence using canonical witnesses; the subsequent monic eliminations themselves are bijections of zero sets.

2.5 Exact degree, not a gate-propagation estimate

In the displayed order the residual degrees are

$$4, 4, 3, 4, 6, 4, 4, 3, 4, 6, 6.$$

Their sum of squares has the following complete degree-twelve homogeneous part, where every supplied unknown has degree one:

$$\boxed{\mathrm{in}_{12}(P_{23}) = 256\beta^4 J^4 u^8 + I^4 f^4 o^4 + \eta^4(-fo + jc + hE)^4.} \tag{7}$$

Indeed the degree-six parts of r_5, r_{10}, r_{11} are respectively $16\beta^2 J^2 u^4$, $I^2 f^2 o^2$, and $-\eta^2(-fo+jc+hE)^2$. Every other residual has lower degree. In particular

$$[\eta^4 j^4 c^4] P_{23} = 1.$$

This proves exact degree twelve after the authentic fixed integers are substituted. It cannot be invalidated by a specialization relation among the coefficient ports.

2.6 The sextic and octic companions

The elimination chain also yields useful lower-degree alternatives. Starting from source 75's centered 34-variable sextic, replace its two native sign guards by the determinant guard. This deletes six private unknowns and introduces three, giving a 31-variable, 18-residual sextic. Its coefficient $[r^2 c^4] = 4$ certifies the degree. Its supplied circuit has 144 operations.

Eliminate $L = q^2 - 1$, $\Delta = a^2 + 4a + 3$, and $B_{\mathrm{aux}} = I^2$. Replace the two remaining positivity representations by the shared certificate while retaining ζ as a coordinate. This gives $31-3-6+4 = 26$ integer unknowns, fourteen residuals, degree eight and 131 operations. The private monomial t^8 has coefficient 1.

Finally eliminate

$$\zeta = 1 - V + jc + hE - \eta, \quad D = 4qu^2 + ac + \gamma(4a+3), \quad V = fo - c.$$

The first elimination makes the shared guard sextic. The other two make the main and auxiliary norms sextic as well, so they cost no further increase in the final degree. These are the three eliminations producing (4). The retained 25-variable intermediate documents this chain, but is dominated by the 23-variable endpoint and is not counted as a separate frontier result.

3 Centered norms: twenty-four and twenty-two integers

Two additional changes improve the variable/degree tradeoff. Both use the integral coordinate shear

$$y_{\mathrm{old}} = fo + y, \qquad y = y_{\mathrm{old}} - fo. \tag{8}$$

This is a bijection on integer coordinates, since f, o are retained. Together with $V = fo - c$, it gives the exact factorization

$$y_{\mathrm{old}}^2 - V^2 = (c + y)(2fo + y - c). \tag{9}$$

The leading square cancels, saving one degree in this difference. These equations use y for the centered root; it no longer denotes the uncentered y of Section 2.

3.1 A degree-ten equation in twenty-four integers

Start with the 26-integer octic described above. Retain ζ as an unknown, set $k = \eta + \zeta$, eliminate $D = X + ac + \gamma H$ and $V = fo - c$, and apply (8). The centered auxiliary residual is

$$A_{\mathrm{aux}} = (fo + y)^2 - I^2(c + y)(2fo + y - c) - 1. \tag{10}$$

Its degree is five. The directly substituted main norm $N = (X + ac + \gamma H)^2 - \Delta c^2 - 1$ still has degree six. A different row representing that norm will lower its degree.

Put $b = ac + \gamma H$ and $Y = a - E$, as before, and define

$$M = XE + 2Eb + Y\big[H(2ac\gamma + H\gamma^2 - c^2) - 1\big]. \tag{11}$$

Since $\Delta = a^2 + H$, direct expansion gives

$$b^2 - \Delta c^2 = H(2ac\gamma + H\gamma^2 - c^2).$$

Consequently the following identity holds as an identity of polynomials:

$$\boxed{M = YN + (X + 2b)(E - XY).} \tag{12}$$

The retained row $E - XY = 0$ therefore makes $M = YN$. The other retained row gives $Y = (2v+1)q^3$. This is never zero at an integer solution: $2v + 1 \neq 0$, and $q = \beta J + 1 \neq 0$ because $\beta > 1$ and J is an integer. Thus replacing $N = 0$ by $M = 0$ is reversible on the common zeros of the other rows. This explicit nonvanishing argument excludes the extra zero component that an arbitrary multiplication could introduce. No positivity, native calibration or Pell theorem is used to establish it.

The unknowns of P_{24} are the twenty-three names in (2), together with ζ; y has the centered meaning. Keep the definitions of $q, C, Y, H, \Delta, X, R, G, \Pi$ from Section 2, but now use $k = \eta + \zeta$, $V = fo - c$, and $b = ac + \gamma H$. Its twelve residuals are

$$\begin{aligned}
s_1 &= Y - (2v + 1)q^3, & s_2 &= E - XY,\\
s_3 &= c - \eta - kY, & s_4 &= \tau^2 + (2\tau - k)E(c - \eta) - 1,\\
s_5 &= M, & s_6 &= k - R - 1 - hE,\\
s_7 &= (K_0 q + X)C - qF - Z(q - 1), & s_8 &= k + V - cj - hE - 1,\\
s_9 &= I - (2r + 1)c^2, & s_{10} &= I^2 - \Delta(f^2 - 1),\\
s_{11} &= A_{\text{aux}}, & s_{12} &= G\eta\zeta - (t^2 - G - \eta\zeta + 1)^2 - \Pi.
\end{aligned} \tag{13}$$

The single equation is

$$\boxed{P_{24} = \sum_{\nu=1}^{12} s_\nu^2 = 0, \qquad \mathbf{x} \in \mathbb{Z}^{24}.} \tag{14}$$

Every residual has degree at most five. The complete degree-ten part is

$$\boxed{\text{in}_{10}(P_{24}) = 64a^4\gamma^2(c + 2\gamma)^2(a - E)^2 + 4I^4 f^2 o^2 (c + y)^2.} \tag{15}$$

These are the leading squares of M and A_{aux}. In particular, $[I^4 f^2 o^2 y^2]P_{24} = 4$, independently of the fixed ports. The supplied circuit has 135 binary arithmetic operations.

Theorem 3.1 (The 24-integer reduction)**.** *The polynomial P_{24} has exactly twenty-four unrestricted integer unknowns and exact degree ten. Its integer solvability is equivalent to that of the 26-integer octic and hence to the pinned finite-refutation predicate. Its unsolvability satisfies (1) for $\mathcal{D} = \mathbb{Z}^{24}$.*

Proof. The monic definitions of D, V have polynomial inverse restorations. The shear (8) is invertible over the integers. Finally (12), together with $E = XY$ and the proved $Y \neq 0$, makes the main-norm replacement reversible. Each step preserves existence of common integer zeros. A sum of integer squares vanishes exactly when all its residuals vanish. The degree certificate (15) proves the remaining claim. □

3.2 A degree-fourteen equation in twenty-two integers

Start with P_{23} rather than P_{24}, so that ζ is again the abbreviation $1 - V + jc + hE - \eta$. Apply the same shear to y. Define

$$Q = \Delta(f^2 - 1), \qquad S = I^2 - Q, \qquad A_{\mathrm{old}} = y_{\mathrm{old}}^2 - I^2(y_{\mathrm{old}}^2 - V^2) - 1.$$

The residual S is already present. Replace A_{old} by

$$A_{\mathrm{new}} = (fo + y)^2 - Q(c + y)(2fo + y - c) - 1. \tag{16}$$

The exact identity

$$A_{\mathrm{new}} = A_{\mathrm{old}} + S(y_{\mathrm{old}}^2 - V^2) \tag{17}$$

shows that this is an invertible elementary row operation in the presence of $S = 0$. It needs no division or nonvanishing assumption.

Now I occurs only in its definition $I = (2r + 1)c^2$ and in S. Eliminate it, deleting its definition row. The strong row becomes

$$S_{22} = (2r + 1)^2 c^4 - \Delta(f^2 - 1). \tag{18}$$

The complete single equation can therefore be written using the already displayed $r_1, \ldots, r_7, r_{11}$ as

$$\boxed{P_{22} = \sum_{\nu=1}^{7} r_\nu^2 + S_{22}^2 + A_{\mathrm{new}}^2 + r_{11}^2 = 0, \qquad \mathbf{x} \in \mathbb{Z}^{22}.} \tag{19}$$

The twenty-two unknowns are precisely (2) with I removed. In the unchanged residuals and abbreviations, $V = fo - c$ and none depends on y, so the centered meaning of y creates no further replacement. The fixed coefficient tuple is unchanged.

The only degree-seven residual is A_{new}. Its leading form is $-2a^2 f^3 o(c + y)$; all other residuals have degree at most six. Hence

$$\boxed{\mathrm{in}_{14}(P_{22}) = 4a^4 f^6 o^2 (c + y)^2, \qquad [a^4 f^6 o^2 y^2] P_{22} = 4.} \tag{20}$$

The degree is exactly fourteen at the actual fixed coefficients. The supplied circuit uses 123 binary operations.

Theorem 3.2 (The 22-integer reduction)**.** *The polynomial P_{22} has twenty-two unrestricted integer unknowns, ten squared residuals, and exact degree fourteen. Its integer solvability is equivalent to that of P_{23} and thus to finite ZFC refutability through the pinned calibration. In particular* (1) *holds for $(P, \mathcal{D}) = (P_{22}, \mathbb{Z}^{22})$.*

Proof. The root shear is an integral bijection. In the common-zero system, (17) has the inverse row operation of subtracting $S(y_{\mathrm{old}}^2 - V^2)$. After this change the monic definition of I can be eliminated, with its displayed integer polynomial as inverse restoration. These steps preserve the entire integer zero set up to the specified coordinate maps. The previous soundness and completeness implications for P_{23} therefore apply. Equation (20) proves exact degree fourteen. □

Direct substitution of $I = (2r + 1)c^2$ into the old auxiliary norm would raise the degree much more. The order matters: remove I from that norm using the strong row, cancel the leading square by centering, and only then eliminate I. The 22-integer result and the degree-ten companion are distinct useful tradeoffs; neither dominates the other in both variable count and degree.

4 Adjacent Pell descent and the eleven-positive-variable equation

The second contribution uses a different coherent tuple, the compact source-44 numerals with $\beta = 2^{590069} - 1$. The same letters K_0, M_C, M_F now refer to that tuple. They are bound separately in `coefficients/ACTIVE_BINDING.json`. The four actual integer payloads and their complete regeneration recipe are included.

4.1 A rational coordinate is already integral

The following descent is source 69's key lemma. Its application here removes a factor previously inserted solely to prove integrality.

Lemma 4.1 (Adjacent-coefficient integrality)**.** *Let $Q > 1$ be an integer. If $V \in \mathbb{Z}$, $y \in \mathbb{Q}$, and*

$$QV^2 - (Q-1)y^2 = 1,$$

then $y \in \mathbb{Z}$.

Proof. Changing signs permits $V, y > 0$; neither is zero. Write $y = b/m$ in lowest terms with $m > 0$. The equation implies $m^2 \mid Q - 1$, hence $m \mid Q - 1$. If $V = 1$ then $y = 1$ and $m = 1$.

For $V > 1$ define

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV.$$

The first is an integer. The identity $(2Q-1)^2 - 4Q(Q-1) = 1$ preserves the equation. Its original form gives $y > V$ and $QV > (Q-1)y$; the latter also follows by squaring positive quantities, since the difference of their squares is $QV^2 + Q - 1 > 0$. These inequalities give $-V < V' < V$. The preserved norm excludes $V' = 0$. Moreover y' has the same reduced denominator m as y because $2Q - 1 \equiv 1 \pmod{m}$. Replace both new coordinates by their absolute values and use induction on the positive integer V. The descent reaches $V = 1$ without changing m, proving $m = 1$. □

The adjacent coefficients and right side 1 matter. For example, $2 \cdot 1^2 - 4(1/2)^2 = 1$ has a nonintegral rational coordinate, so no general statement for arbitrary positive coefficients is being used.

4.2 The exact eleven-variable polynomial

Supply precisely

$$j, F, \alpha, i, T, X, Y, n, \eta, \zeta, \gamma \in \mathbb{Z}_{>0}.$$

All symbols in the following displays are polynomial abbreviations:

$$\begin{aligned}
J &= 3j + 1, & q &= \beta J + 1, & C &= q - F - \alpha,\\
E &= XY, & k &= \eta + \zeta, & c &= kY + \eta,\\
a &= E + Y, & H &= 4a + 3, & \Delta &= a^2 + H,\\
\ell &= EkY, & I &= ic^2, & D_\tau &= \ell(\ell + k) + 1.
\end{aligned}$$

The removal of $q(q-1)$ from I is the structural change from source 70. Define

$$\begin{aligned}
R &= \big((q-1)(q-F)+\alpha\big)(q^2-1)+(M_C+qM_F)J, \\
\widehat{R} &= q^3(q+F+\alpha), & M &= q^3(q-1), \\
B_t &= (qK_0+X)C-qF, & D_f &= 1+\Delta I^2, \\
Q &= \Delta(D_f-1)=\Delta^2 I^2, & \delta &= Q-1, \\
V &= D_f(cT-R)-c, & \overline{V} &= D_f(cT+\widehat{R}), \\
L &= 3\overline{V}, & W &= 1+(q-1)L, \\
\overline{Z} &= M(\ell+k+n+L)+(q-1)Y-q^3WB_t.
\end{aligned}$$

For the retained main carrier put

$$\begin{aligned}
v_0 &= k-R-1, & S &= 4qY, & U &= qS, \\
D_* &= U(X+ac)+H(U\gamma+SX-v_0), \\
A_1 &= D_*^2-U^2(1+\Delta c^2).
\end{aligned}$$

For the three-root norm set

$$\begin{aligned}
A &= M^2D_\tau, & B_f &= M^2D_f, & C_n &= M^2(QV^2-1), \\
T_n &= \delta^2 AB_f, & R_n &= \delta C_n, \\
J_n &= \delta(\overline{Z}^2-A-B_f)-C_n, \\
N_c &= J_n^2-4\big(T_n+R_n(A+B_f)\big), \\
N_8 &= N_c^2-64T_nR_n\overline{Z}^2.
\end{aligned}$$

The polynomial is

$$\boxed{P_{11}=N_8+A_1+N_8A_1=0.} \tag{21}$$

This literal division-free circuit has 111 binary operations.

4.3 Radical recovery without a prime-support multiplier

The universal identity

$$\begin{aligned}
F_8(z;a,b,c) &= \big((z^2-a-b-c)^2-4(ab+ac+bc)\big)^2-64abcz^2 \\
&= \prod_{\varepsilon\in\{\pm1\}^3}(z-\varepsilon_1\sqrt{a}-\varepsilon_2\sqrt{b}-\varepsilon_3\sqrt{c})
\end{aligned}$$

is a polynomial identity. For positive rational radicands all eight displayed roots are real; it does not require independent square classes.

We recall source 70's elementary radical lemma. If three positive rational radicands are pairwise distinct and no signed sum of their square roots is zero, any rational signed sum of them makes each root rational. Indeed let signed roots x_1,x_2,x_3 have nonzero rational sum z, and write $a_i=x_i^2$. The quantities

$$u=\frac{z^2-a_1-a_2-a_3}{2}, \qquad v=\frac{u^2-a_1a_2-a_1a_3-a_2a_3}{2z}$$

are rational and equal to $\sum_{i<j}x_ix_j$ and $x_1x_2x_3$. Then

$$x_1=\frac{a_1(a_1+u)}{a_1z+v}, \qquad a_1z+v=x_1(x_1+x_2)(x_1+x_3)\neq 0.$$

The other two formulas are cyclic. Distinct radicands exclude the two possible cancellations in each denominator.

Here the three radicands before multiplication by M^2 are

$$D_\tau, \qquad D_f, \qquad D_y = \frac{QV^2 - 1}{Q - 1}.$$

All of the following bounds hold for arbitrary positive supplied inputs, before any equation or the inequality $C > 0$ has been assumed. In particular $c \geq 3$, $a \geq 2$, $\Delta \geq 15$, $I \geq c^2$, and

$$1 < D_\tau < D_f < Q.$$

Put $b = cT - R$. If $b = 0$, $V = -c$; otherwise $|V| \geq D_f - c > 1$. Thus $|V| > 1$ in all cases and $D_y = V^2 + (V^2 - 1)/(Q - 1) > V^2$. If D_y equalled either integer $D \in \{D_\tau, D_f\}$, then $Q(D - V^2) = D - 1$, impossible because $0 < D - 1 < Q$.

Writing $t_\tau = \sqrt{D_\tau}$ and $f = \sqrt{D_f}$ gives

$$t_\tau < (a + 1)c, \qquad f > (a + 1)c^2 \geq 3c^2.$$

When $b = 0$, $\sqrt{D_y} < c + 1 < f - t_\tau$. When $b \neq 0$, $\sqrt{D_y} > D_f - c > 2f > f + t_\tau$. One root therefore dominates the sum of the other two, excluding a zero signed sum.

The authentic mask bounds give $0 < M_C < \beta$ and $0 < M_F < 2\beta$. They imply $|R| < \widehat{R}$ for every positive input. For example

$$|R| < q^4 + Fq^3 + (\alpha + 2)q^2 < q^3(q + F + \alpha),$$

using $q \geq 16$, $\alpha \geq 1$. The positive integer gap, together with $D_f > c$, gives $|V| < \overline{V}$. Hence

$$f < \overline{V}, \qquad \sqrt{D_y} < 2|V| < 2\overline{V}, \qquad L > f + \sqrt{D_y}, \quad L > k.$$

These are exactly the estimates needed for source 70's root transport; none uses the removed factor $q(q - 1)$ in I.

At a zero of N_8, the identity

$$N_8 = \delta^4 F_8(\overline{Z}; M^2 D_\tau, M^2 D_f, M^2 D_y)$$

supplies signed roots with $\overline{Z}/M = \tau_* + f_* + y_*$. The radical lemma makes them rational. The first two are integral because their squares are integers. The third satisfies

$$QV^2 - (Q - 1)y_*^2 = 1.$$

Lemma 4.1 therefore gives $y_* \in \mathbb{Z}$ directly. This replaces source 70's prime-support argument and is why the stronger divisibility of I is unnecessary.

4.4 Soundness and constructive completeness

For soundness recover

$$f_{\text{old}} = |f_*|, \quad u_{\text{old}} = |y_*| - V, \quad r_{\text{old}} = n + L - f_* - y_*, \quad i_{\text{old}} = i.$$

The inequalities above make $u_{\text{old}}, r_{\text{old}}$ strictly positive. The signed roots must be used in r_{old}; replacing them there by absolute values would change the transport identity. With

$$Z = M(\ell + k + r_{\text{old}}) + (q - 1)Y - q^3 W B_t$$

one obtains $Z = M\tau_*$. The four source-70 weighted-carrier equations are now

$$Z^2 - M^2 D_\tau = 0, \qquad A_1 = 0,$$
$$f_{\mathrm{old}}^2 - \Delta(i_{\mathrm{old}}c^2)^2 - 1 = 0,$$
$$(V + u_{\mathrm{old}})^2 - \Delta(f_{\mathrm{old}}^2 - 1)u_{\mathrm{old}}(2V + u_{\mathrm{old}}) - 1 = 0.$$

Its theorem applies to every positive integer transport weight, here L. It restores the normalized positive source and hence a finite refutation.

For clarity, this inherited weighted-carrier theorem has a specific order of recovery. First $M \mid Z$ yields $q^3 \mid Y$ and $(q-1) \mid B_t$, and the positive transport gap gives $B_t > 0$, thus $C > 0$. Only then are the native bounds $3q < R < q^4 - 2$ used. The adaptive main carrier gives a positive main Pell root, the strong/auxiliary equations recover its index, and its large modulus recovers $q \mid X$ and the deleted index quotient. This order avoids using native bounds before their hypotheses have been established.

Conversely choose canonical normalized source witnesses for a finite refutation. A positive Pell pair with ordinate divisible by c^2 is available; it follows either from the canonical source or from a positive power of the Pell matrix that is the identity modulo c^2. Thus $I = ic^2$ with $i > 0$. The actual auxiliary reconstruction lemma of source 61 applies to every positive such pair with $c \mid I$, canonical $R \geq 3$, $R \equiv 3 \pmod 4$, and $c = \psi_{a+2}(R)$, supplying positive T, u and $y = V + u$. Here $\psi_A(R)$ is the coefficient of $\sqrt{A^2-1}$ in $(A + \sqrt{A^2-1})^R$. Choose these before computing L, W. Canonical transport has $t_0 = B_t/(q-1) \geq q$, $s_0 = Y/q^3 < k$, and $\ell < \tau < \ell + k$. Its positive gap can therefore be set to

$$r = Wt_0 + \tau - \ell - k - s_0 > qW - 2k > L.$$

Now $n = r - L + f + y > 0$ and $\overline{Z} = M(\tau + f + y)$, so $N_8 = 0$. The adaptive main-carrier witness is supplied by the same canonical construction. This proves completeness with no extra strong-index congruence.

Finally N_8 is a square modulo 4, and so is A_1 because $4 \mid U$. The equation $P_{11} = 0$ is $(1 + N_8)(1 + A_1) = 1$. The negative-unit case would require both residuals to be -2, impossible modulo 4. Thus this single equation is equivalent to their simultaneous vanishing.

4.5 Exact degree and its coefficient certificate

The exact degrees are

register	I	D_f	Q	V	L	$\overline{Z}$	N_8	A_1
degree	5	14	18	18	18	24	264	13.

The degree-fourteen terms in A_1 cancel; the exact audit checks this identity rather than trusting naive gate degrees. Writing $q_0 = 3\beta j$ and $k_0 = \eta + \zeta$, the leading forms satisfy

$$\mathrm{in}_{264}(N_8) = Q_0^4 \overline{Z}_0^8,$$

where

$$Q_0 = i^2 k_0^4 X^4 Y^8,$$
$$B_{t,0} = (K_0 q_0 + X)(q_0 - F - \alpha) - q_0 F,$$
$$\overline{Z}_0 = -3i^2 q_0^7 k_0^4 X^2 Y^6 (q_0 + F + \alpha) B_{t,0},$$
$$\mathrm{in}_{13}(A_1) = 32 q_0^2 k_0 X^2 Y^4 (4q_0^2 Y\gamma + q_0^3(q_0 - F)).$$

In particular the degree-277 monomial has coefficient

$$[j^{60}i^{24}\eta^{49}X^{42}Y^{85}\alpha^{16}\gamma]P_{11} = 2^7\,3^8(3\beta)^{60} > 0.$$

The degree falls by 48, or about 14.77%, from source 70's 325. The live arithmetic count falls by two multiplications, from 113 to 111.

5 A twelve-positive-variable companion of degree 78

This construction composes source 69's norm merger with source 70's adaptive main carrier. Supply

$$j, F, \alpha, t, T, X, Y, r, \eta, \zeta, n, \gamma \in \mathbb{Z}_{>0}.$$

Use $J = 6j + 1$, $q = \beta J + 1$, and the same compact tuple and common registers $C, E, k, c, a, H, \Delta, \ell, R, M, B_t$ as above. Set

$$\begin{aligned} W &= (q-1)k + 1, \\ Z &= M(\ell + k + r) + (q-1)Y - q^3 W B_t, \\ A_0 &= Z^2 - M^2\big(\ell(\ell+k) + 1\big). \end{aligned}$$

Use the adaptive $S = 4qY$, $U = qS$, D_* and A_1 from the preceding section, in place of source 69's scales q^4, q^5. The remaining registers are

$$\begin{aligned} f &= t + 1, & \rho &= f^2 - 1, & d &= \Delta c^4, \quad Q = \Delta\rho, \\ V &= f^2(cT - R) - c, \\ h(z) &= (1-Q)z^2 + 2(1-Q)Vz + V^2 - 1, \\ b_N &= 2(1-Q)(n+V), & A_N &= d\,h(n) + (1-Q)\rho, \\ N &= A_N^2 - d\,b_N^2\rho. \end{aligned}$$

The single equation is

$$\boxed{P_{12} = (1 + A_0)(1 + A_1)(1 + N) - 1 = 0.} \tag{22}$$

5.1 The merger and its exact recovery

Assume $d, \rho > 0$, $Q > 1$, $V > 1$, $n > 0$ and $N = 0$. Then $b_N \neq 0$. Define the proof-only rational numbers

$$i_* = \frac{A_N}{d b_N}, \qquad u_* = n - i_*.$$

One gets $di_*^2 = \rho$ and $h(u_*) = 0$. Writing $y_* = V + u_*$, the latter equation is $QV^2 - (Q-1)y_*^2 = 1$. Lemma 4.1 gives $y_* \in \mathbb{Z}$, hence $u_*, i_* \in \mathbb{Z}$. Recover the positive witnesses $i = |i_*|$ and $u = |y_*| - V$; the second is positive because $|y_*| > |V|$. Conversely, if $di^2 = \rho$ and $h(u) = 0$, set $n = i + u$. The identity

$$N\big|_{\rho = di^2} = d^2 h(n-i)h(n+i)$$

shows $N = 0$. This is an exact polynomial identity; it is not an unjustified rational-root resultant inference.

The required hypotheses precede this recovery. The equations $A_0 = A_1 = 0$ alone give $C > 0$, $q^3 \mid Y$ and positive first and main Pell indices. The first index is at least two; otherwise $k = 2$, $\eta = \zeta = 1$ would contradict the smallest possible main ordinate. Therefore

$$c > kY > 8XY^3 \geq 8q^9 > 2R.$$

Since $f = t + 1 \geq 2$, this gives $V = f^2(cT - R) - c > c > 1$. The adaptive carrier has the same hypotheses and rank consequences here as in source 70. In particular $q \equiv 2 \pmod 3$ and $q^3 \mid Y$ make its relevant gcd at most 3, while its remaining modulus exceeds $R + 1$. Thus replacing q^5, q^4 by $4q^2Y, 4qY$ does not weaken the source reconstruction.

The recovered i, u restore the four positive source residuals, with transport weight k. Conversely, canonical witnesses have $J \equiv 1 \pmod 6$ because the radix exponent and repunit length are odd; they supply $t = f - 1 > 0$ and $n = i + u > 0$. The first transport and adaptive main-carrier gaps are positive by the canonical construction. This proves the existence equivalence with the same native source.

The radix q is even, $4 \mid U$, and b_N is even. Hence all three residuals A_0, A_1, N are squares modulo 4. Their unit-product finalizer in (22) is equivalent to their separate vanishing: any factor -1 would again require a residual -2.

5.2 Degree and arithmetic cost

The exact residual degrees are $15, 13, 50$, so the product has degree 78. An explicit coefficient surviving fixed-port specialization is

$$[j^{18}F^2\eta^{11}X^{12}Y^{23}t^8n^2\gamma\alpha]P_{12} = 1024(6\beta)^{18} > 0.$$

The new scale computation costs one more live gate than the old scale, giving 103 operations. This is a degree improvement from 82 to 78, about 4.88%, and an explicit tradeoff rather than domination of the old circuit in every measure.

6 A separate compiler improvement: the modulus supplies the index

The results above improve complete equations in the existing native-compiler family. This section concerns a different source architecture. It removes one existential coordinate from the inert-prime CRT certificate of source 75. Its factorial, binomial and variable-exponent operations have not been compiled and counted as a new ordinary polynomial. Thus it is a proved compiler lemma, not another claimed Con(ZFC) endpoint.

6.1 The homogeneous substitution

Fix an explicit polynomial $R(i, w_1, \ldots, w_m; \mathbf{x}) \in \mathbb{Z}[i, \mathbf{w}, \mathbf{x}]$, where $m \geq 1$, and let R_{abs} replace each coefficient by its absolute value. The external parameters $\mathbf{x}$ are natural numbers. For $n \geq 1$ and $U \geq 0$, put

$$H = R_{\mathrm{abs}}(n, U, \ldots, U; \mathbf{x}) + n + U + 1, \qquad s = 4H!,$$

$$m_i = s(i + 1) - 1 \quad (0 \leq i < n), \qquad B = \prod_{i=0}^{n-1} m_i.$$

If an arithmetic DAG already represents $R = e - f$ with nonnegative coefficient polynomials, the majorant $e + f$ can be used instead, preserving that sharing. Write $R = \sum_{a=0}^{d} R_a(\mathbf{w}; \mathbf{x}) i^a$, with $d = 0$ for the zero polynomial, and define the ordinary integer polynomial

$$\widehat{R}(s, \mathbf{w}; \mathbf{x}) = \sum_{a=0}^{d} R_a(\mathbf{w}; \mathbf{x})(1 - s)^a s^{d-a}. \tag{23}$$

The right side is a finite polynomial definition. The notation $s^d R((1 - s)/s, \mathbf{w}; \mathbf{x})$ explains the construction, but does not introduce division into its polynomial representation.

For a fixed binary tree, define a norm recursively by $\mathcal{N}(z) = z$ and $\mathcal{N}(L, R) = \mathcal{N}(L)^2 + \mathcal{N}(R)^2$. Use the convention $\binom{A}{k} = 0$ when $0 \le A < k$.

Homogeneous-index closure theorem. For each fixed $n, U, \mathbf{x}$ as above, the bounded assertion

$$\forall i < n \; \exists \mathbf{w} \in \{0, \dots, U\}^m \quad R(i, \mathbf{w}; \mathbf{x}) = 0$$

is equivalent to the existence of natural $A_1, \dots, A_m$ such that

$$B \mid \mathcal{N}\left(\binom{A_1}{U+1}, \dots, \binom{A_m}{U+1}, \widehat{R}(s, \mathbf{A}; \mathbf{x})\right). \tag{24}$$

There is no separately encoded index and no index-congruence leaf.

Proof. Every prime divisor of m_i exceeds H, because a prime at most H divides $H!$ and makes $m_i \equiv -1$. The m_i are pairwise coprime: a common prime divisor of m_i, m_j would divide $j - i$, since it cannot divide s, but $0 < |j - i| < n \le H$. Each m_i is $3 \pmod 4$, so it has a prime divisor $p_i \equiv 3 \pmod 4$. Over $\mathbb{F}_{p_i}$, a sum of two squares vanishes only when both entries vanish. Induction on the norm tree therefore makes vanishing of its value equivalent to vanishing of all its leaves.

Suppose (24) holds. For each i, all leaves vanish modulo p_i. The identity

$$(U+1)!\binom{A_j}{U+1} = \prod_{r=0}^{U}(A_j - r)$$

and $p_i > H > U + 1$ give $A_j \equiv w_j \pmod{p_i}$ for some $0 \le w_j \le U$. The same prime works for all coordinates. Furthermore $s(i+1) \equiv 1 \pmod{p_i}$, so

$$\widehat{R}(s, \mathbf{A}; \mathbf{x}) \equiv s^d R(i, \mathbf{w}; \mathbf{x}) \pmod{p_i}.$$

Since s is invertible modulo p_i, the residual is zero modulo p_i. Its absolute value is at most $R_{\mathrm{abs}}(n, U, \dots, U; \mathbf{x}) < H < p_i$, so it is zero in $\mathbb{Z}$.

Conversely, select bounded witnesses for every $i < n$. CRT gives natural A_j with $A_j \equiv w_{ij} \pmod{m_i}$. Every m_i is coprime to $(U+1)!$, so all binomial leaves vanish modulo m_i. The same homogeneous identity and the local equality show that the residual leaf does also. Pairwise coprimality then proves (24). □

The proof is finite arithmetic formalizable in ZFC; it uses no consistency or arithmetic-soundness assumption. Existentially quantifying U recovers unbounded witnesses at the finitely many indices. Since the norm has at least two leaves, it is nonnegative, and its divisibility can be written as one equality $\mathcal{N} = Bv$ with $v \in \mathbb{N}$.

6.2 The finite product and the exact saving

As in source 75, the product defining B has the special-function expression

$$Q = (sn)^n + 1, \qquad M = sQ + 1,$$
$$B = s^n n! \binom{Q+n}{n} \bmod M.$$

Indeed the numerator is $\prod_{j=1}^{n} s(Q+j)$, whose factors reduce to $sj-1$. Their positive product is below $(sn)^n < Q < M$, so its least remainder is the required product. The variable exponent and the special functions remain genuine compilation obligations.

Source 75 used the leaves

$$\binom{A_j}{U+1} \ (1 \le j \le m), \qquad R(c, \mathbf{A}; \mathbf{x}), \qquad s(c+1) - 1.$$

Equation (23) removes the existential coordinate c, rather than merely dropping its check. The number of leaves falls from $m+2$ to $m+1$. A balanced norm tree with k leaves has maximum degree $2^{\lceil \log_2 k \rceil}$ in its leaf symbols. When $m = 2^r - 1$, $r \ge 1$, this bound falls from 2^{r+1} to 2^r. This is not an unconditional degree reduction after substitution and compilation: homogenization changes the degree in s, and all graph witnesses and shared operations still need to be counted.

More generally, a modulus of the form $si + t$, with s a unit modulo the selected prime divisor, supplies the residue $i = -t/s$. The corresponding finite homogeneous polynomial uses $(-t)^a s^{d-a}$. This is a reusable way of recovering an index from the modulus itself.

6.3 Checks and limits

The authored standard-library checker verifies 1,125 exact homogeneous identities, 650 modular index identities, 35 product and coprimality cases, 156,584 finite-field norm evaluations, and 896 full CRT fixtures. Of the last set, 79 satisfy the certificate, 594 fail a range condition, and 223 fail the local residual. Four negative controls check why one needs an inert prime, the nested norm, a residual bound and an invertible scale. The tests run with ordinary and optimized Python and contain no assertions that optimization could suppress. The universal theorem rests on the written proof, not these finite diagnostics.

This component does not instantiate a new proof-checker source, eliminate the special-function graphs, or establish a new minimum-variable consistency equation independent of K215. Its exact contribution is the one-coordinate, one-leaf improvement to a separately identified compiler stage.

7 Reproducibility and interpretation of the checks

The top-level command `python3 verify.py` checks the delivered polynomials, their fixed numerals, and their regeneration. It requires only Python's standard library. The PDF is compiled from this LaTeX source; the executable arithmetic does not depend on a LaTeX installation.

The integer audit reconstructs the compact eleven residuals independently, restores the arithmetic rows of the signed forty-variable source, compares them as exact integer polynomials, and expands the final sum of squares. It finds 16,360 monomials when the fixed ports are temporarily left as symbolic coefficient parameters. These are not floating-point tests. An independently written audit also verifies the exact degree certificate after coefficient specialization: its selected coefficient is the numeral 1, so no relation among the fixed ports can cancel it. A separate independent extension

audit verifies the centered equations, the complete leading forms of P_{24} and P_{22}, and their source restorations. It expands respectively 5,634 and 16,525 symbolic-port monomials and finds the displayed coefficient 4 in each leading form.

The positive audits check every named cut in the arithmetic circuits, the universal norm identities, the cancellation in the main Pell residual, and nonzero leading coefficients on the authentic fixed-coefficient slice. The eleven-variable circuit has 65 multiplications, 29 additions and 17 subtractions. The twelve-variable circuit has 58 multiplications, 26 additions and 19 subtractions. Squares and scalar products are charged; fixed leaves and reference reuse are free. Replacing a register by a polynomial abbreviation does not create another quantified unknown.

The positive coefficient recipe is replayed from the literal transition table and the native finite language. Missing duplicate source files in the placed upstream package were restored byte-for-byte using their pinned hashes, so the delivered replay works without the original checkout. The integer component ships all six actual coefficient payloads and their bindings, including its two determinant scales. The two components use different coherent native tuples; their ports must never be mixed.

Finite diagnostic examples exercise the Pell descent, the shared sign guard, and the CRT theorem, including negative controls. They support the implementation. They cannot prove ZFC consistent, establish the absence of all roots, or replace any universal theorem. The mathematical proof uses the written existence implications; the checks establish exact finite data and algebraic implementation facts.

8 Further research questions

1. **Can the 22-integer equation be lowered to degree twelve?** The centered strong-norm row operation already eliminates $I = (2r+1)c^2$ at degree fourteen. Its remaining bottleneck is the degree-seven auxiliary residual, whose leading term is $-2a^2f^3o(c+y)$. A further relation that reduces this term could improve the endpoint without restoring a coordinate. Alternatively, can another coordinate be removed while retaining degree fourteen?

2. **Can the determinant be recalibrated on a smaller native tuple?** The current integer construction inherits the source-34 radix $2^{1064921}$. The positive constructions use 2^{590069}. Reusing smaller ports requires a whole-tuple canonical-witness proof of the determinant margin and the odd-rank promises; replacing one numeral in isolation is invalid.

3. **Can the eleven-variable norm be centered at its auxiliary root without creating a sign loop?** The bound on $f+y$ drives the degree of the transport coordinate. A bound on $f+(y-V)$ would be much smaller when $V>0$, but soundness must establish the needed sign before using the transport to prove $C>0$. Removing that dependence loop would be a substantive further improvement.

4. **Can the CRT index lemma produce a complete alternative source?** Instantiate it with an explicit proof checker, compile factorial, binomial, power and remainder graphs, count all witnesses and operations, and prove the actual syntactic ZFC calibration. Only then can it be compared with the finished native equations.

5. **Can the ordinary proof be internalized modularly?** The polynomial coordinate maps, the strict-positivity equivalence and the adjacent-Pell descent are small, separate targets. Their internal ZFC derivations should precede the larger native compiler calibration. The relevant endpoint must quantify over all internal naturals, not only external numeral instances.

6. **Which complexity measure should govern further search?** The variables/degree pair favors P_{22} and P_{23} over the older 24-variable degree-1936 tradeoff. The fewest-variable objective still favors the high-degree nine-positive or ten-integer forms. A reproducible search should report the domain, degree, live operations and fixed-coefficient size together, and preserve these distinct objectives.

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